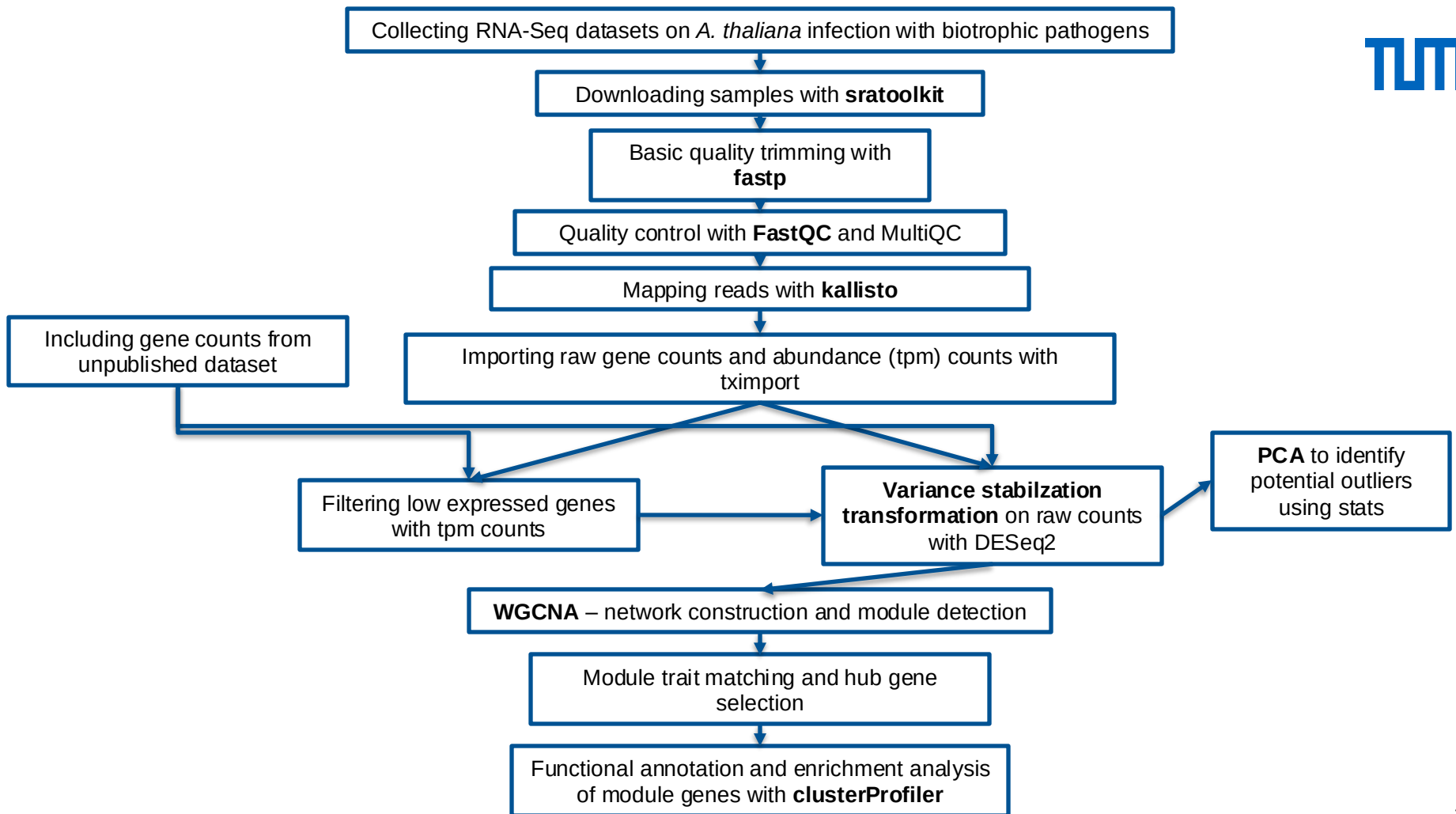


Unveiling Commonalities and Divergences in Biotrophic Plant-Microbe Interactions using Gene Co-Expression Networks in *Arabidopsis thaliana*

Bachelor thesis

Luisa Bender

Freising, 10. März 2025



About the pathogens

- Fungal biotrophs (powdery mildew disease)
- Oomycete biotroph (downy mildew)
- Root and cyst nematodes
- Protist biotroph
- Viral biotrophs

Control samples:

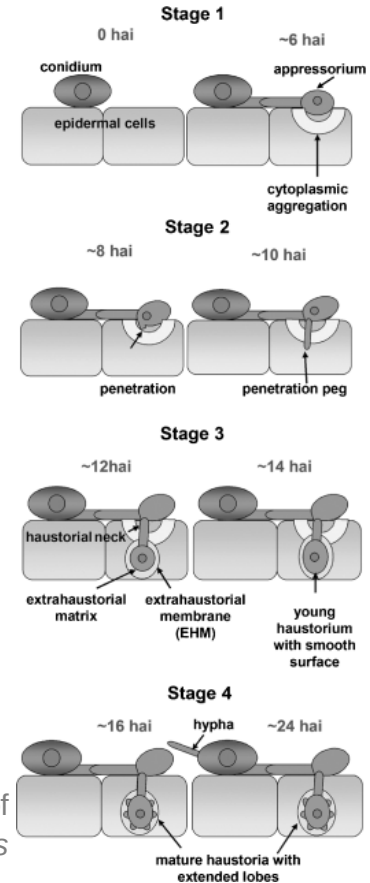
- Necrotrophic pathogens
- Drought

Fungal biotrophs

- *Golovinomyces orontii*
- *Erysiphe cichoracearum*
- *Erysiphe cruciferarum*
- Causing foliar **powdery mildew disease**
- Powdery mildews rely on living plant tissue for growth and reproduction
- Life cycle begins with landing of a conidiospore on the leaf surface
- Formation of a **haustorium** → serves as „feeding“ organ for nutrient uptake



Arabidopsis infected with *G. orontii*



Infection stages of *E. cichoracearum* on *Arabidopsis*

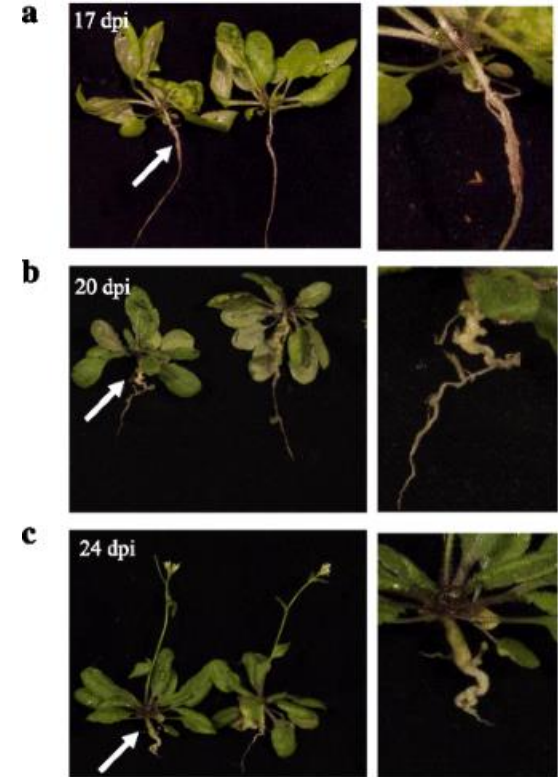
Oomycete biotroph

- *Hyaloperonospora arabidopsidis*
- causing **downy mildew** disease
- shares common infection process with fungi causing powdery mildew disease
 - formation of haustoria
- Oomycetes are phylogenetically close brown algae (distinct from true fungi)
- Obligate biotrophs → cannot live without their host



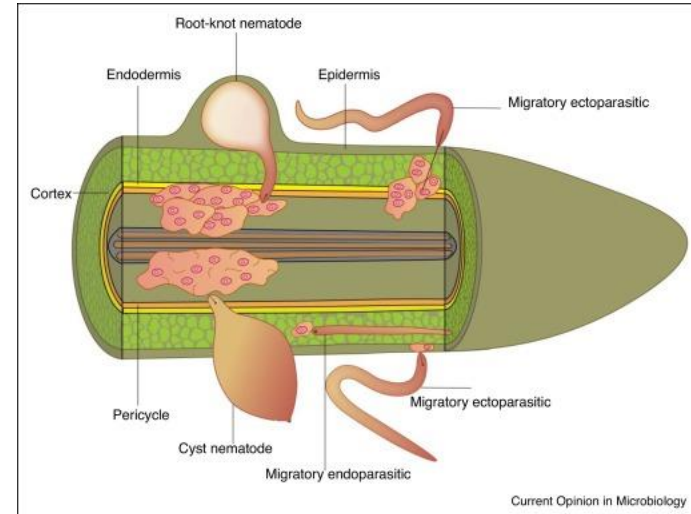
Protist biotroph

- *Plasmodiophora brassicae*
 - Clubroot pathogen
 - Modulates host root development
 - Initiating formation of nutrient-replete **galls** within the root systems
 - Root infection negatively affects the uptake of water and nutrients in host plants
- Reducing growth and productivity



Nematodes

- endoparasitic
- Cyst nematode (CN): *Heterodora schachtii*
- Root-knot nematode (RKN): *Meloidogyne javanica*
- Invasion of plant root
- Move through different tissue layers to reach vascular cylinder
- Formation of feedings sites:
 - CN: syncytium (fusion of cells)
 - RKN: several giant cells
- uptake nutrients from the plant
- Induce pathologically disturbed allocation of photosynthetic products that reduces plant growth and yield



Viral biotrophs

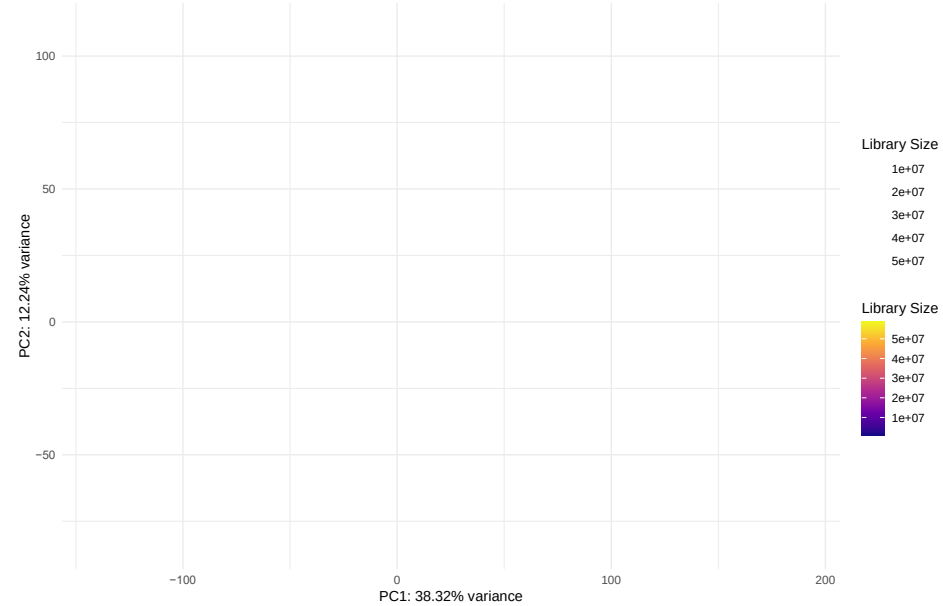
- *Turnip mosaic virus (TuMV)* – potyvirus
- Enters cells and begins replication in the cytoplasm
- forming viral replication complexes on the ER
- uses host translation machinery to produce viral proteins
- Moves intracellularly to reach plasmodesmata
- Eventually reaching the phloem allowing systemic infection throughout the plant

- *Turnip yellows virus (TuYV)* – polerovirus
- Replicates only in phloem companion cells
- More specialized than TuMV

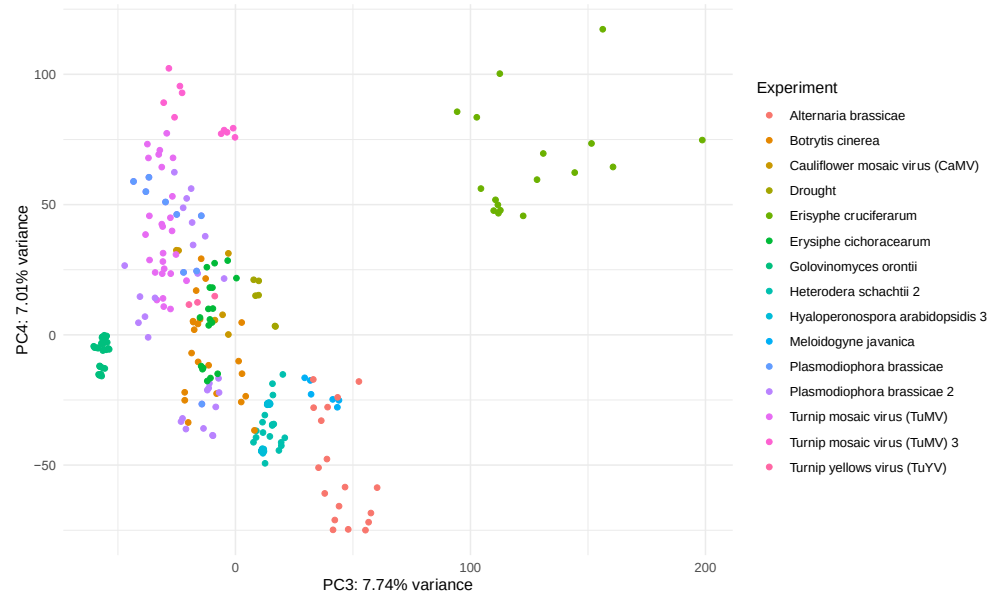
PCA of filtered vst counts



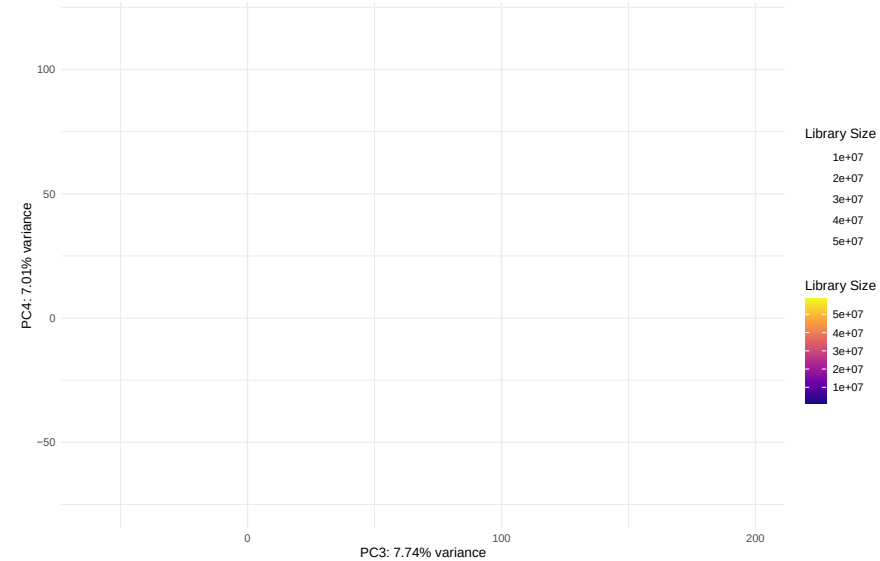
PCA with Library Size

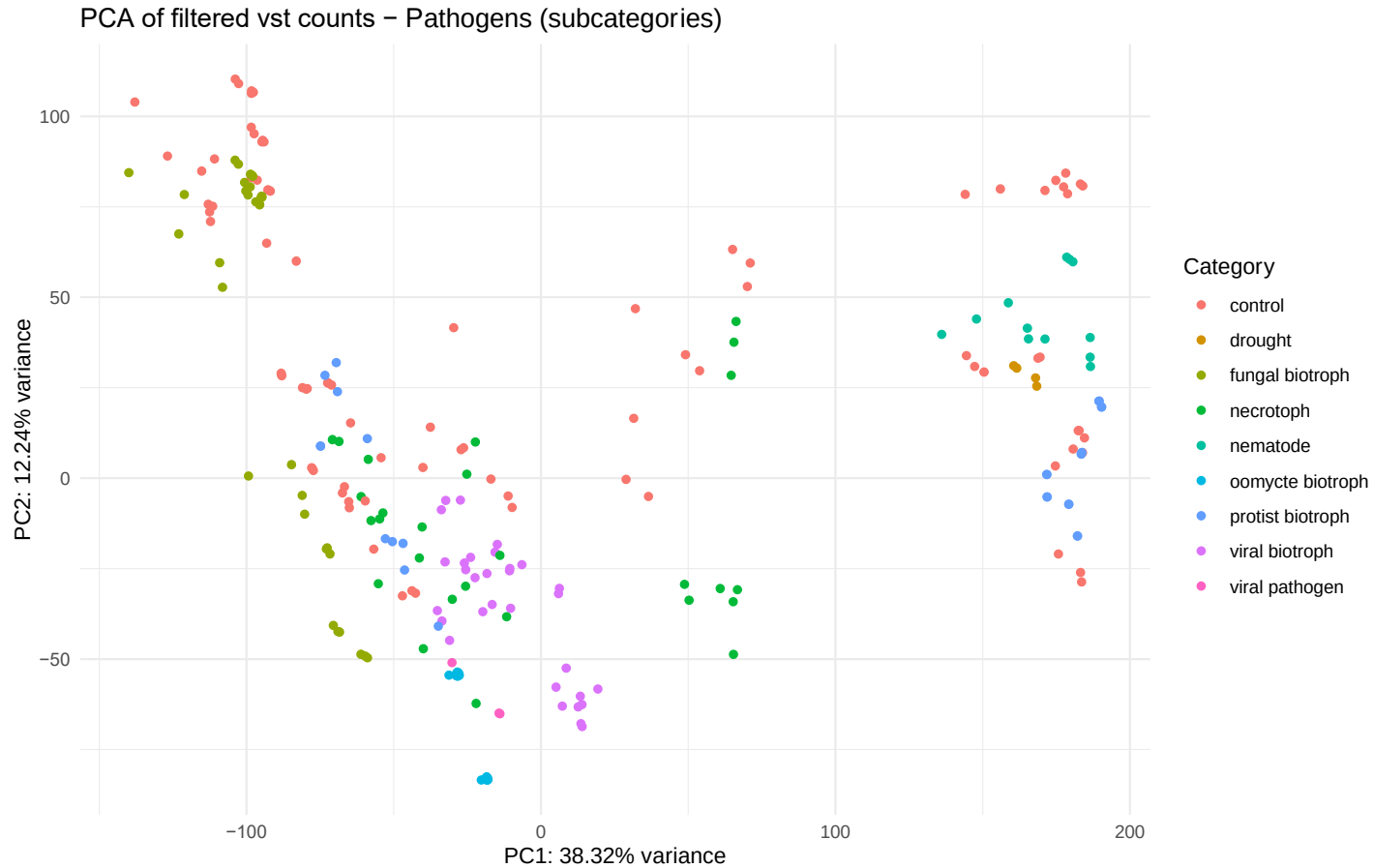


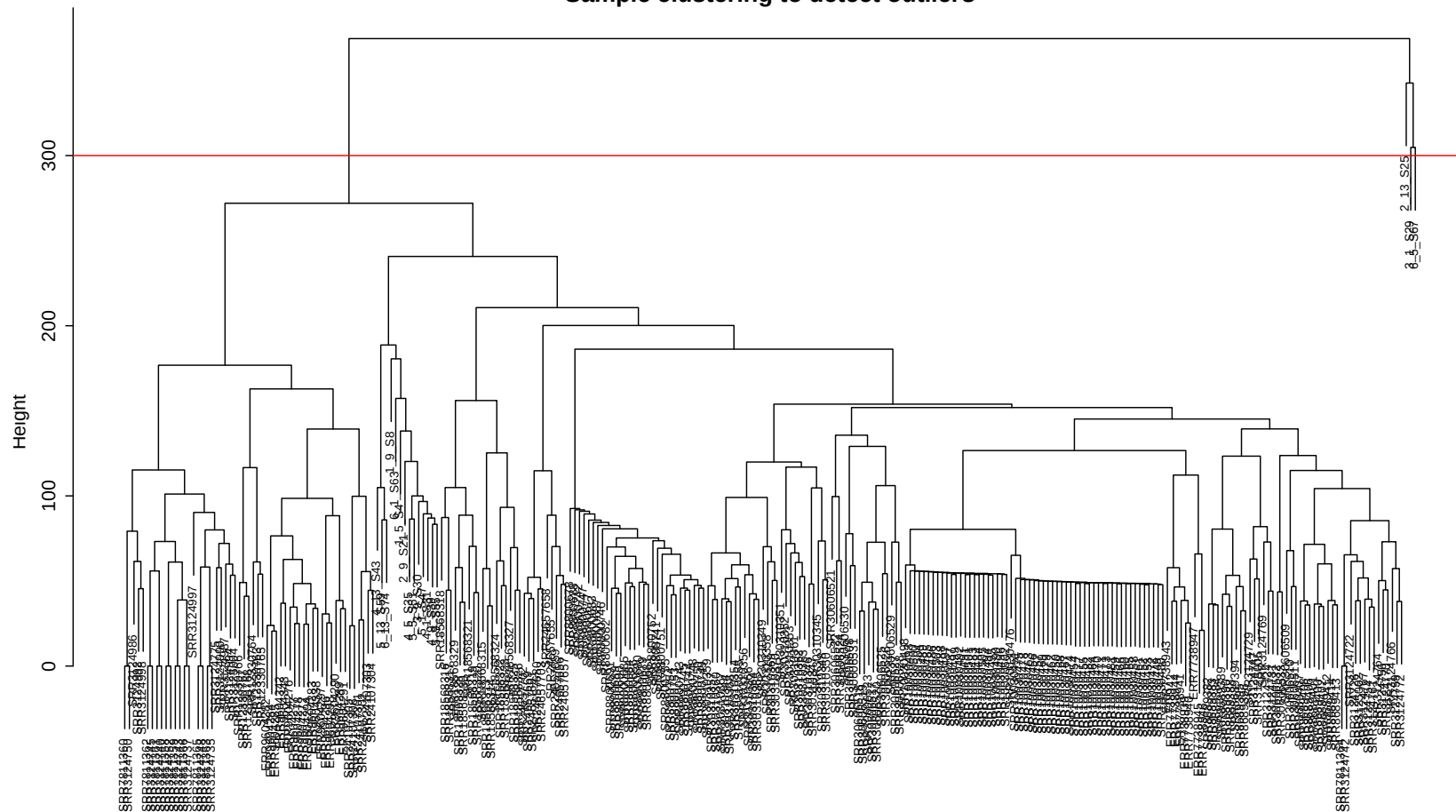
PCA of filtered vst counts



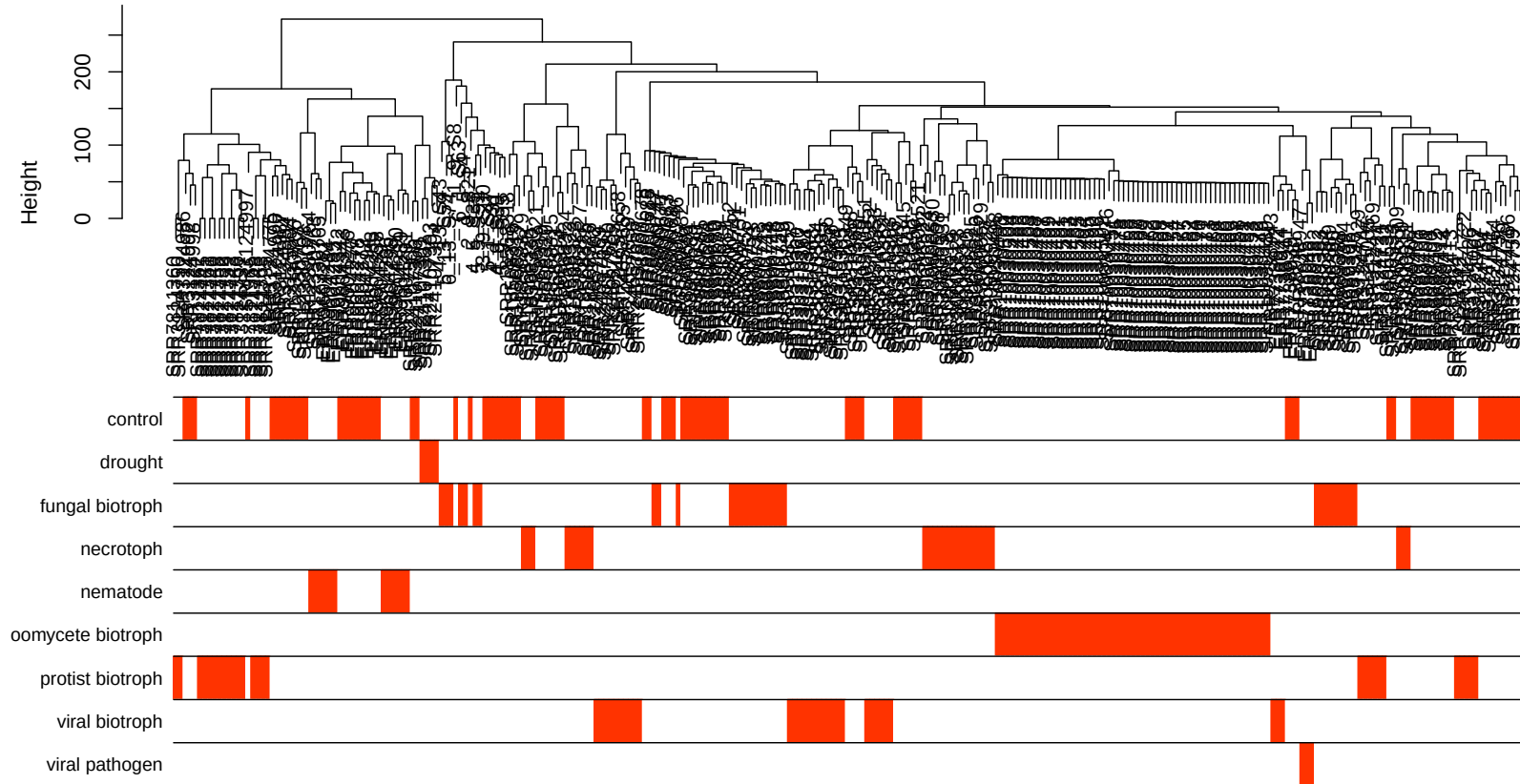
PCA with Library Size



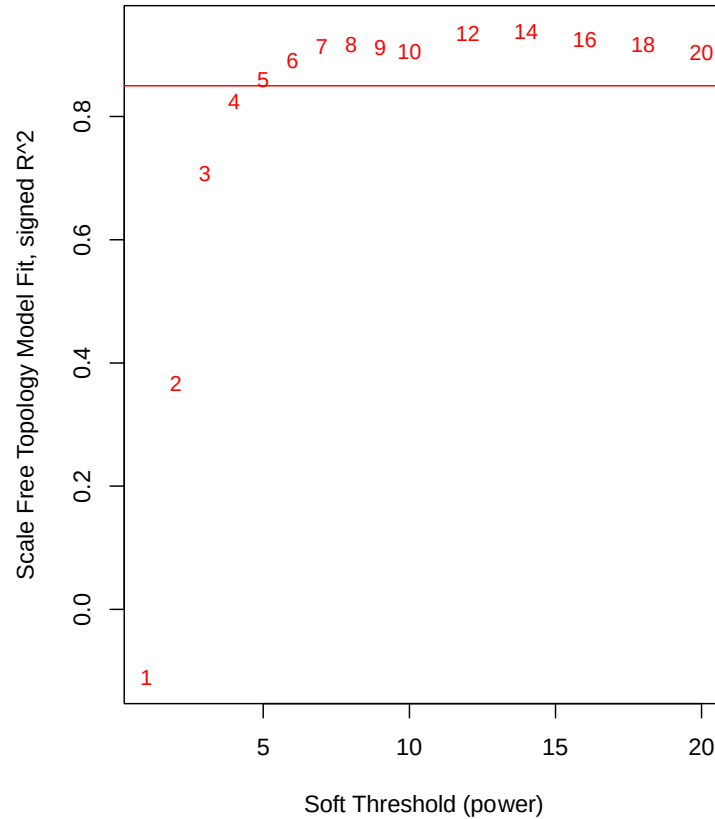




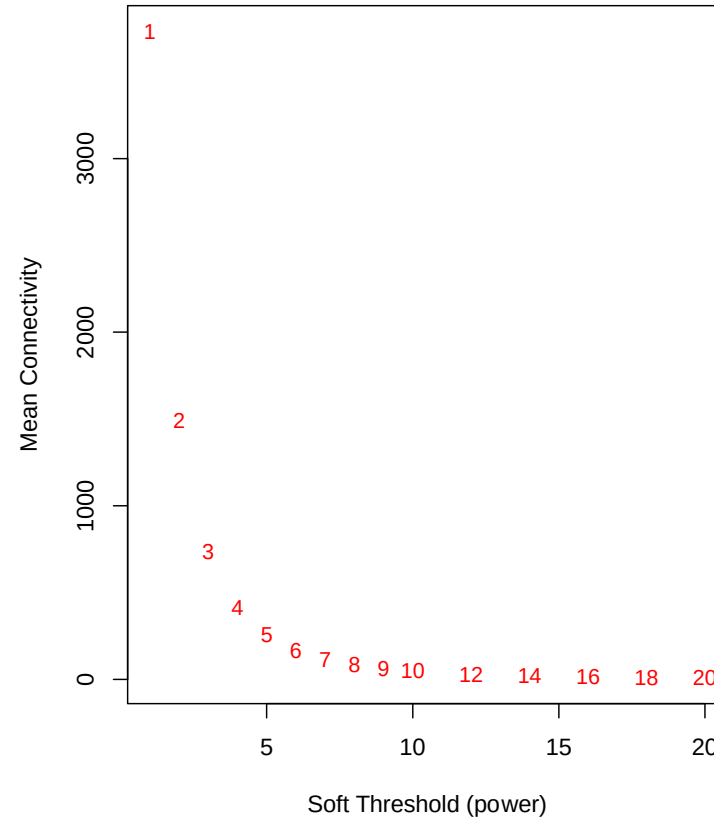
Sample dendrogram and metadata heatmap



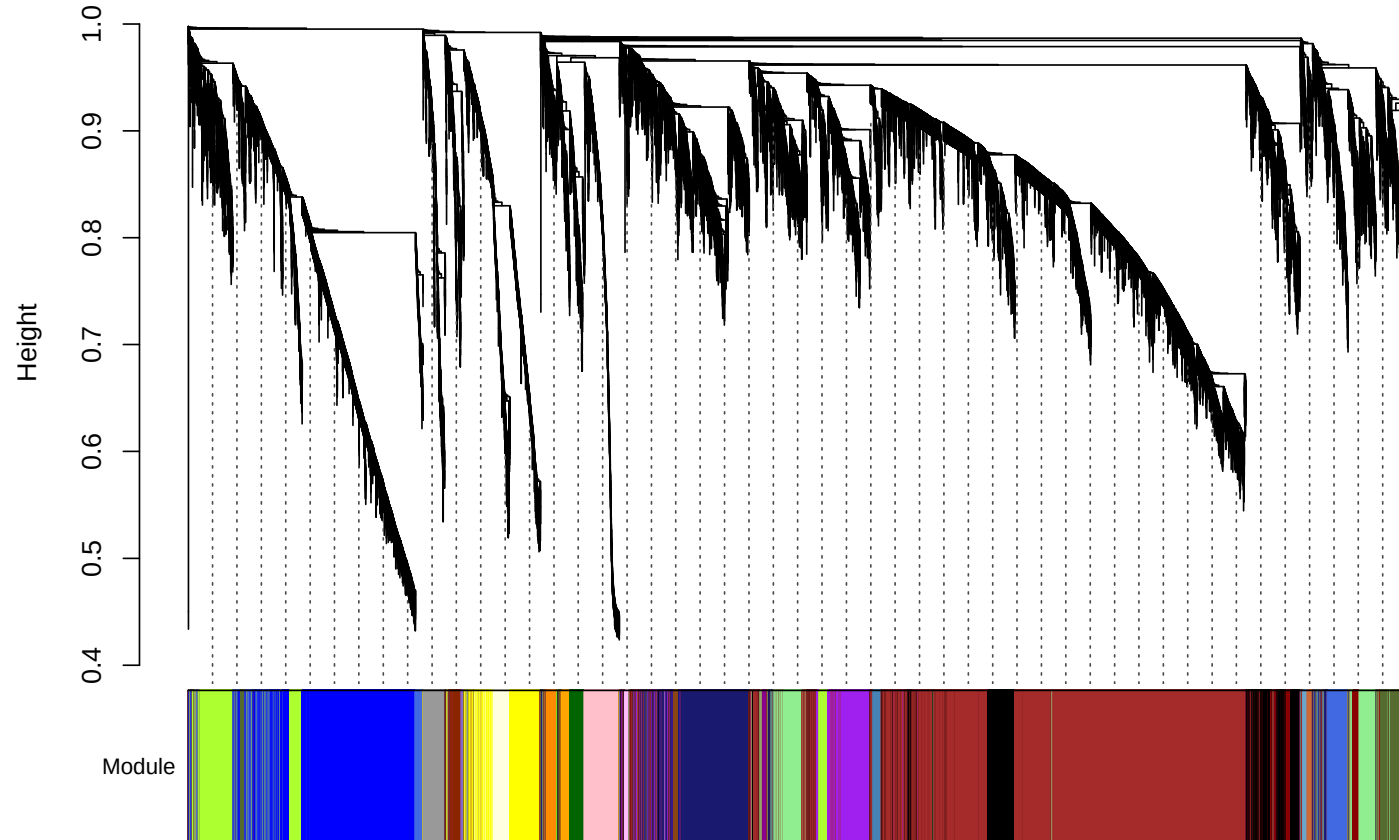
Scale independence



Mean connectivity

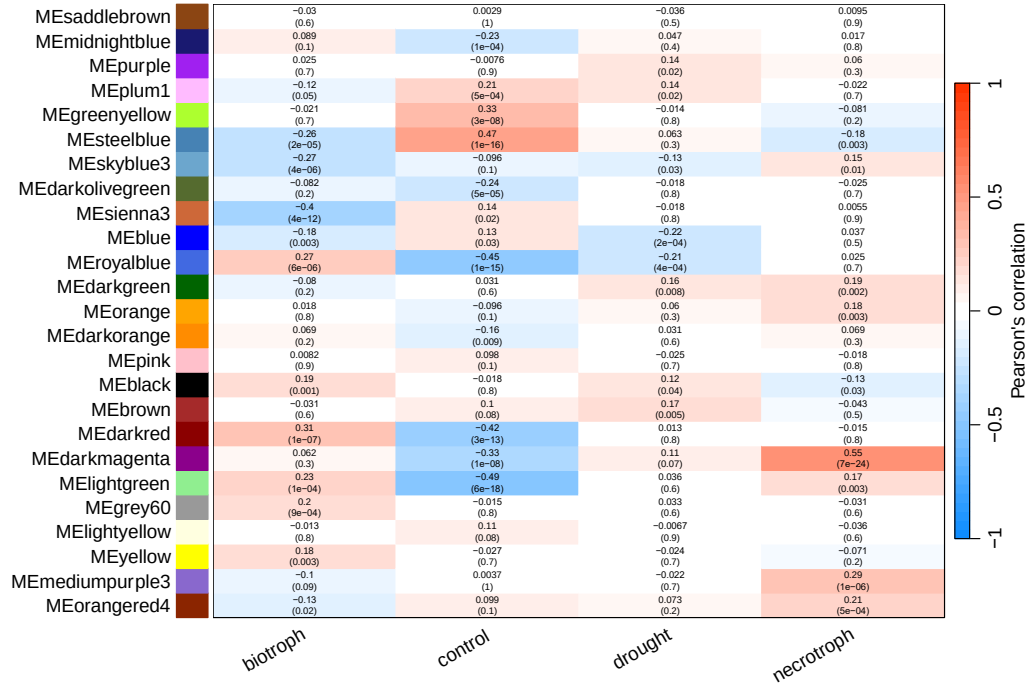


Gene dendrogram and module colors

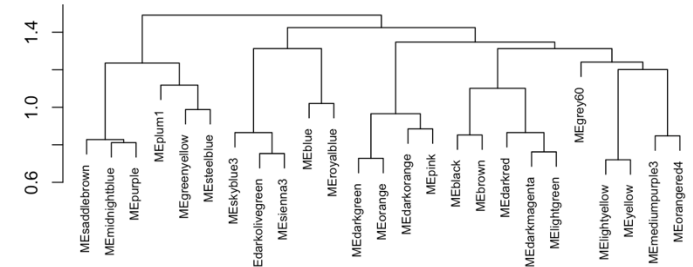


25 Modules
No grey module

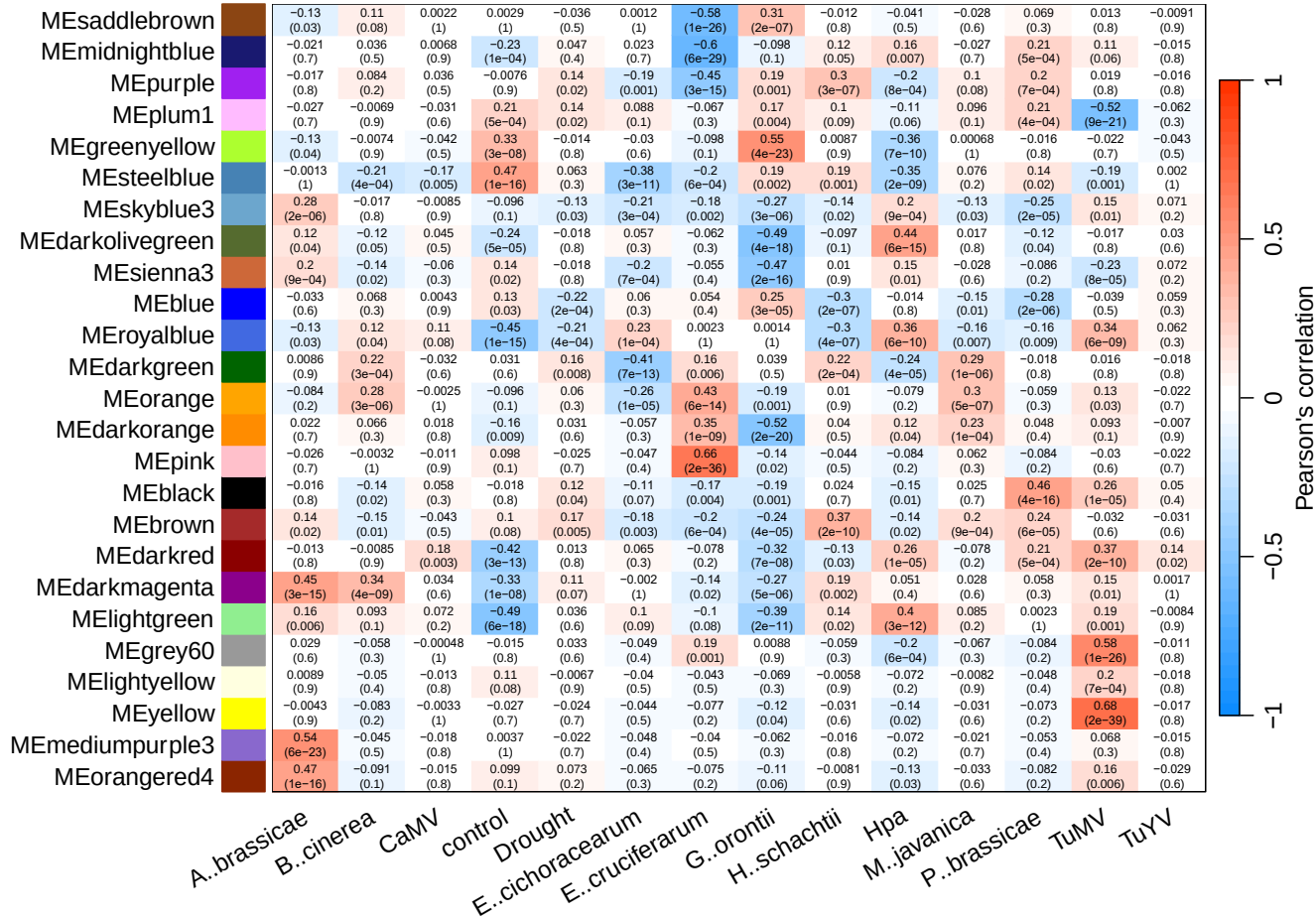
Module-trait relationships – Pathogens (categories)



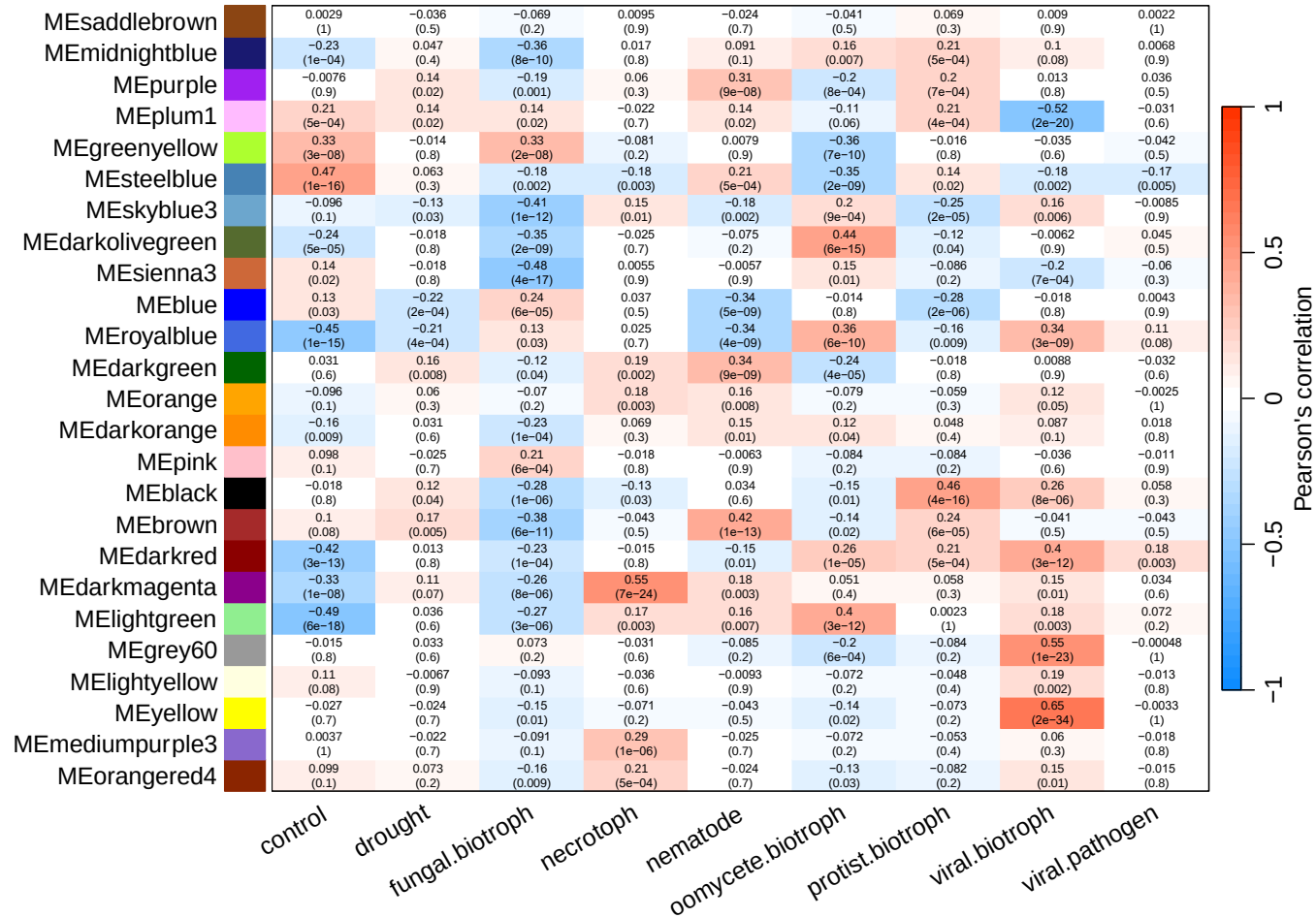
Hierarchical Clustering of Module Eigengenes



Module-trait relationships – Pathogens



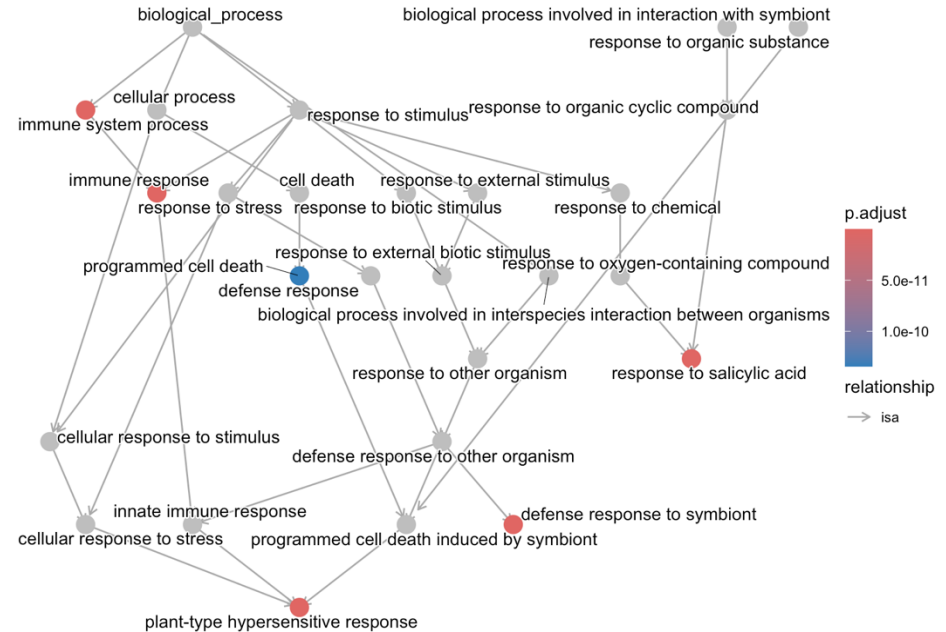
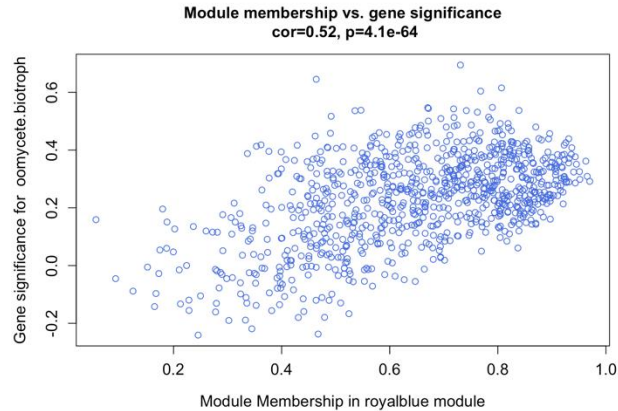
Module-trait relationships – Pathogens (subcategories)



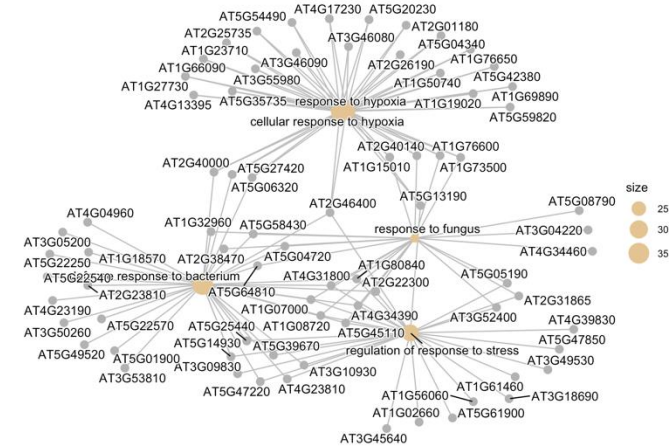
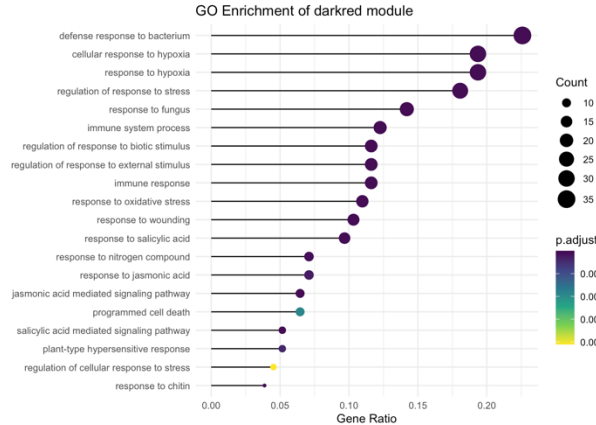
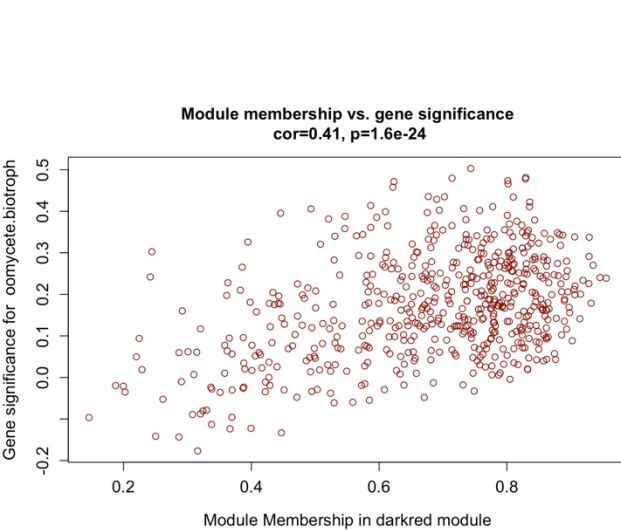
Methods for analyzing modules

- Module Membership (MM) / kME (signedKME()): correlation between each gene and each module eigengene
→ describes the connectivity of a gene inside a module
- Gene Significance (GS): correlation between each gene and a given trait
→ Helps identify genes with a high significance for a pathogen infection
- Hub gene: important gene that has a high connectivity inside a module
- Genes with **kME > 0.8** considered as hub genes
- Functional enrichment of modules and hub genes with clusterProfiler
 - Gene Ontology (GO) overrepresentation analysis, mostly biological processes
 - KEGG pathway enrichment

Oomycete biotroph - *Hyaloperonospora arabidopsidis*



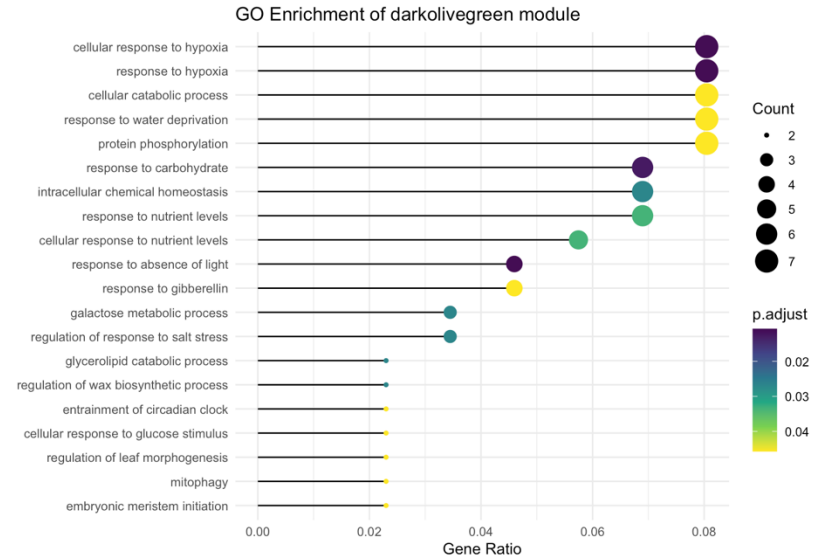
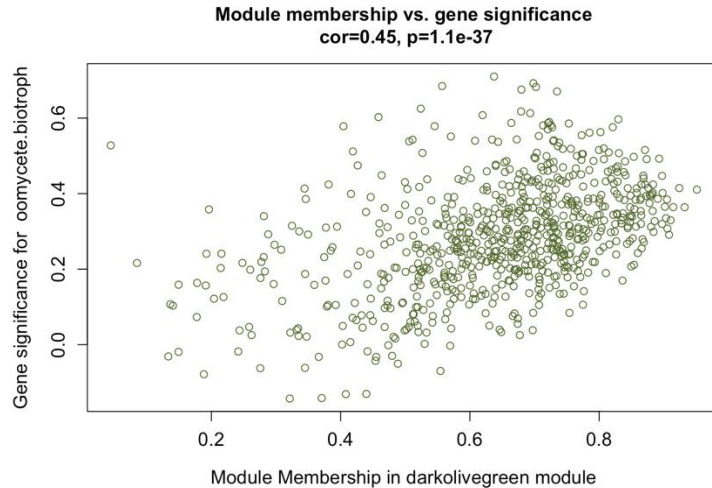
Oomycete biotroph - *Hyaloperonospora arabidopsidis*



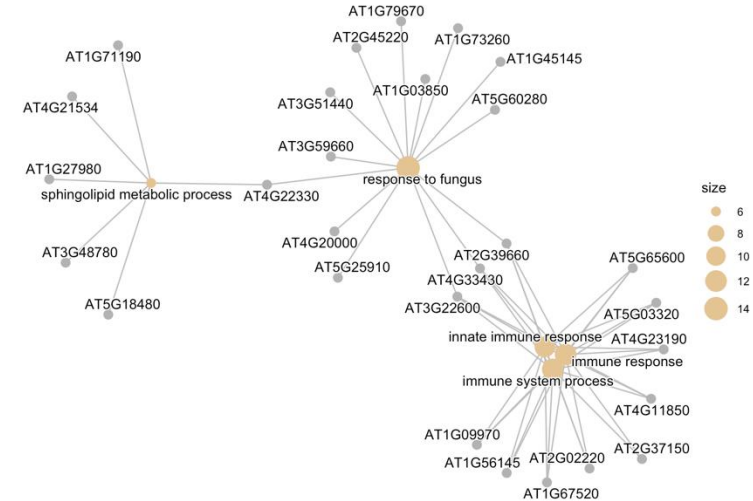
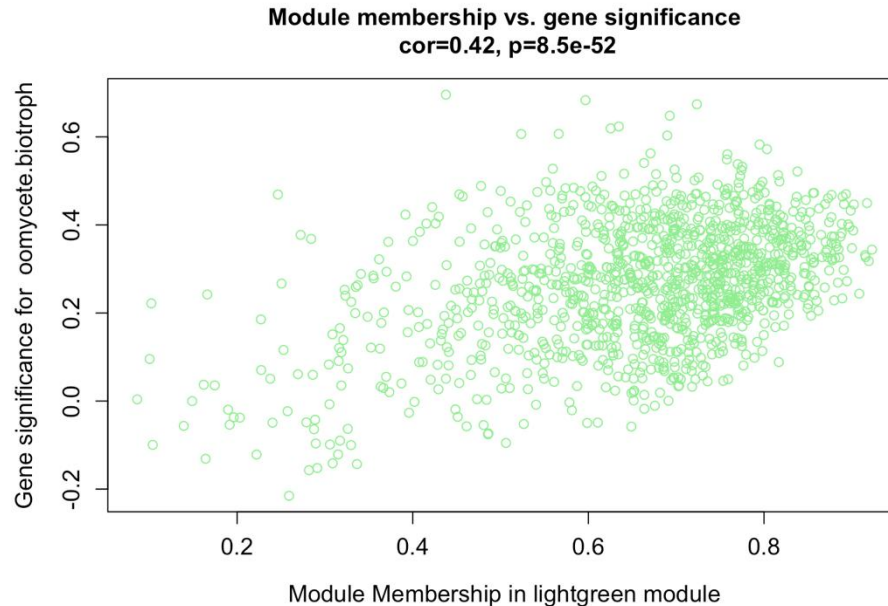
darkred: enriched in plant-pathogen interaction, MAPK signaling pathway, Glycerolipid metabolism

Single-gene multifunctions: genes could play roles in hypoxia or pathogen responses at the corresponding stress → WRKYs

Oomycete biotroph - *Hyaloperonospora arabidopsidis*



Oomycete biotroph - *Hyaloperonospora arabidopsidis*



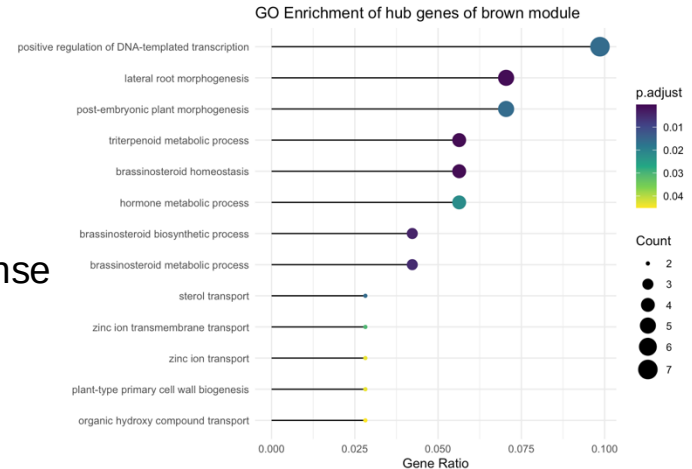
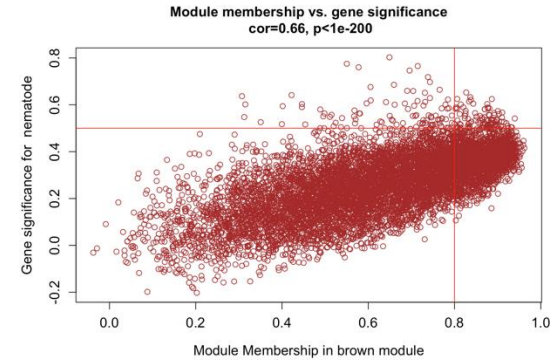
Nematodes – *M. javanica* and *H. schachtii*

brown module:

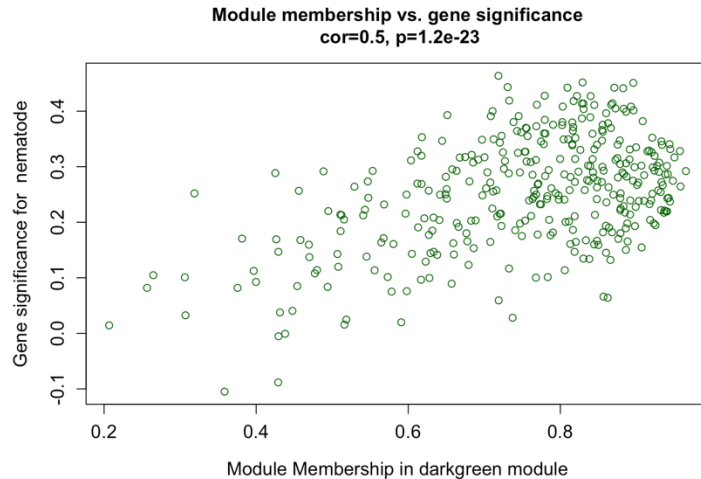
→ enriched in root morphogenesis, plant-type cell wall organization, plant epidermis development

Significant for nematodes: positive regulation of DNA-templates transcription and lateral root morphogenesis

- Nematodes manipulate lateral root development to create feeding structures (syncytia)
- Secrete effectors that modulate host transcription factors
- Plant may activate its own transcriptional networks in response to stress and infection



Nematodes – *M. javanica* and *H. schachtii*



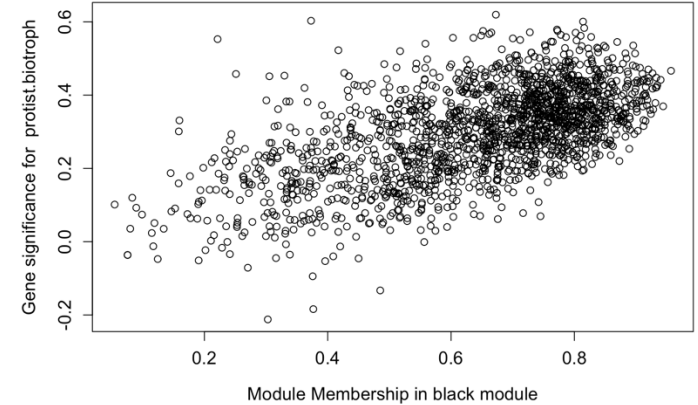
darkgreen: enriched in translation, ribosome biogenesis, cytoplasmic translation, rRNA processing

- Increased protein synthesis demand
- Often associated with rapid growth, stress responses or metabolic shifts
- For nematodes: could suggest that the nematode is inducing high protein production to maintain feeding sites

Protist biotroph – *P. brassicae*

- enriched in plant-pathogen interaction pathway
- hypoxia response**: contributes positively to clubroot development
- jasmonic acid signaling**: part of defense response
- hub genes also belong to JAZ or TIFY family
 - 13 JAZ genes have been reported in *A.thaliana*
 - participate in plant development and defense reactions

Module membership vs. gene significance
cor=0.6, p=1.9e-174



Hypoxia response in *Arabidopsis* roots infected by *Plasmodiophora brassicae* supports the development of clubroot



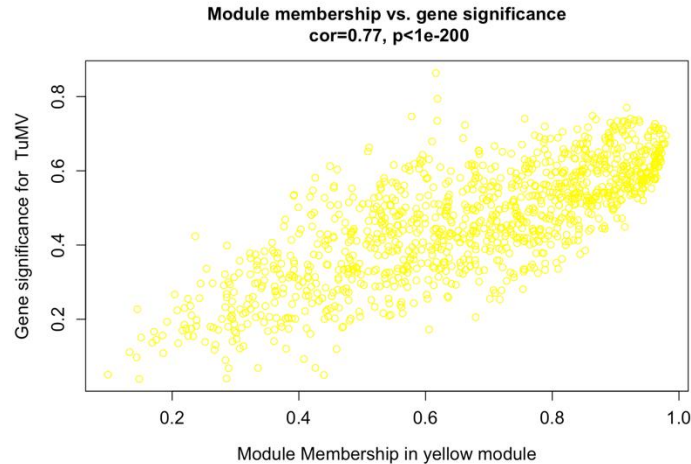
Antoine Grivot¹*, Gautier Richard¹, Tanguy Lime¹, Séverine Lemarié¹, Mélanie Jubault¹, Christine Lariagon¹, Jocelyne Lemoine¹, Jorge Vicente², Alexandre Robert-Seilaniantz¹, Michael J. Holdsworth³ and Maria J. Manzaneres-Dauleux¹

Article

Genome-Wide Identification and Variation Analysis of JAZ Family Reveals *BnaJAZ8.C03* Involved in the Resistance to *Plasmodiophora brassicae* in *Brassica napus*

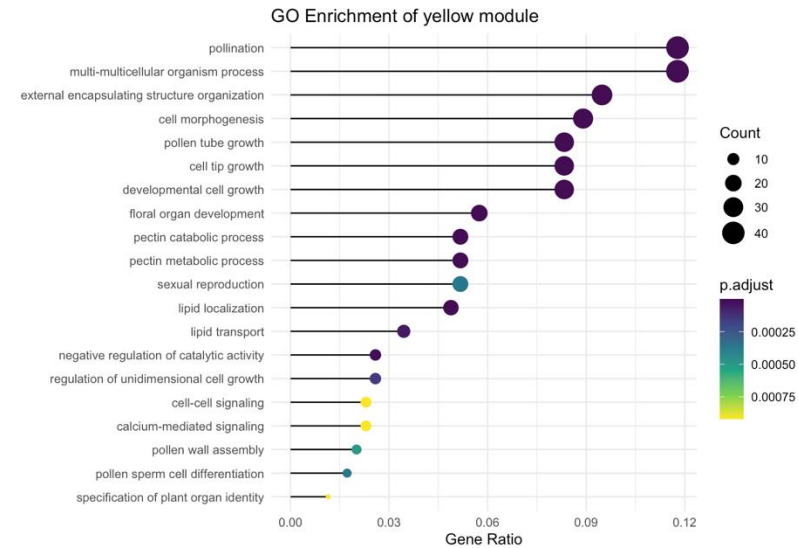
Lixia Li[†], Gaoxiang Ji[†], Wenjie Guan, Fang Qian, Hao Li, Guangqin Cai^{*} and Xiaoming Wu^{*}

Viral biotroph – Turnip mosaic virus



Enriched in following pathways:

- Pentose and glucuronate interconversion
- Cutin, suberine and wax biosynthesis



Viral biotroph – Turnip mosaic virus

- Very specific pathways and biological processes
→ could suggest changes in cell wall and reproductive development of the plant
- Cutin, suberine and wax are components of cell wall and protective barriers
→ Defense mechanism of the plant trying to build mechanical barriers against the invasion of the virus

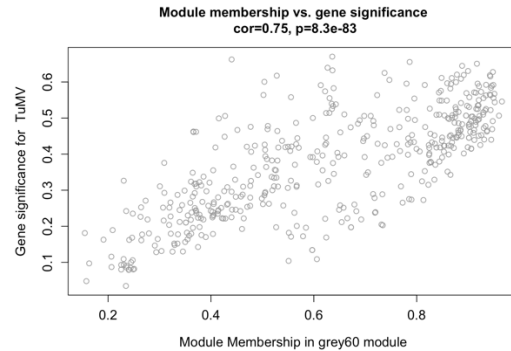
> [Front Plant Sci.](#) 2021 Oct 8;12:741050. doi: 10.3389/fpls.2021.741050. eCollection 2021.

The Effects of *Turnip Mosaic Virus* Infections on the Deposition of Secondary Cell Walls and Developmental Defects in Arabidopsis Plants Are Virus-Strain Specific

Silvia López-González ¹, Concepción Gómez-Mena ², Flora Sánchez ¹,
Mathias Schuetz ³, A Lacey Samuels ³, Fernando Ponz ¹

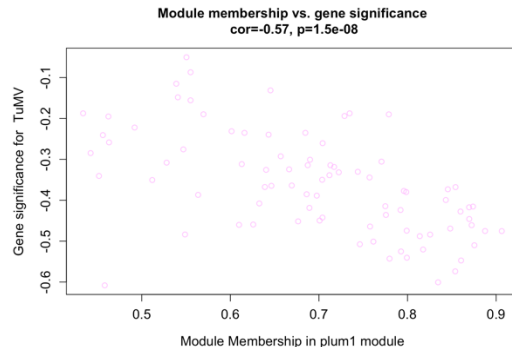
- Viral strains of TuMV disrupt secondary cell wall deposition
- Interfere with plant development
- Developmental alterations could maybe lead to dysregulation of reproductive pathways

Viral biotroph – Turnip mosaic virus



Grey60: enriched in oxidative phosphorylation, purine nucleoside triphosphate biosynthetic process

- Indicate changes in mitochondrial energy production
- Manipulation of host cellular metabolism



Plum1: enriched in DNA damage response, double strand break repair

Fungal biotrophs

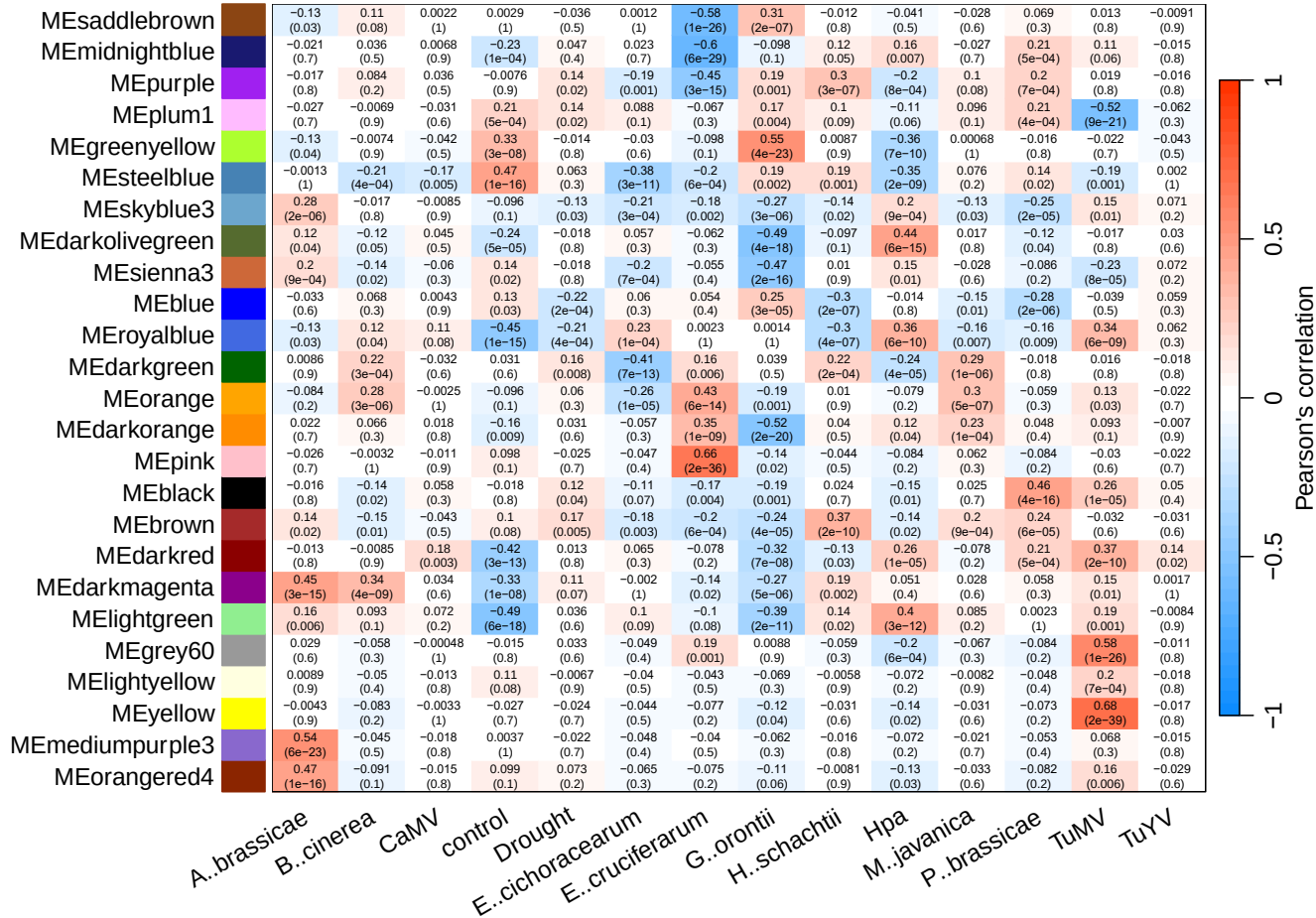
- Different modules correlating with all three fungal biotrophs
 - Maybe different infection strategies?
 - Genetic differences between fungal species?

G.orontii

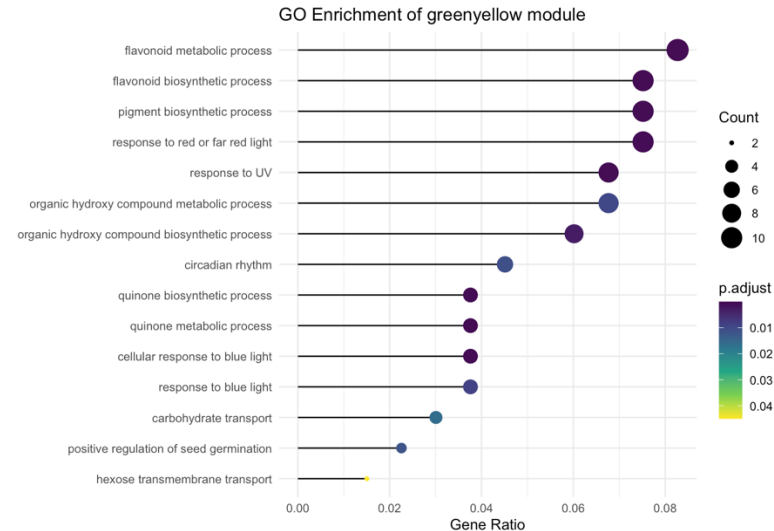
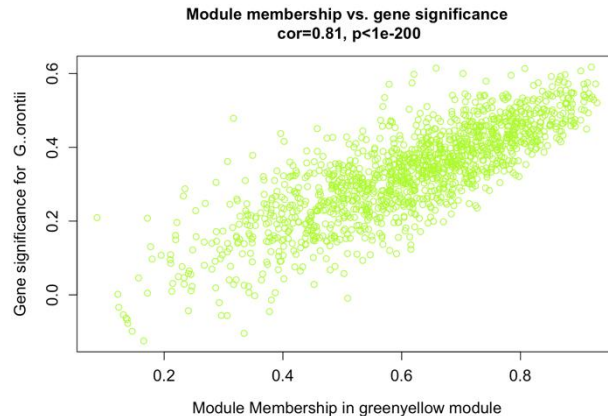
E.cichoracearum

E.cruciferarum

Module-trait relationships – Pathogens



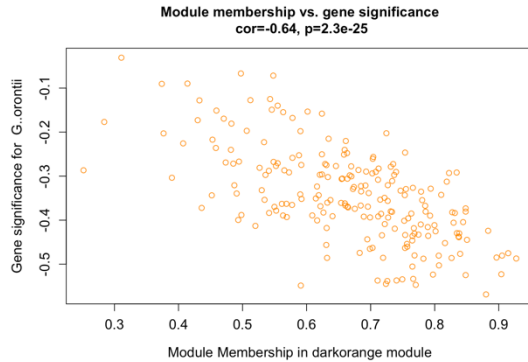
Golovinomyces orontii



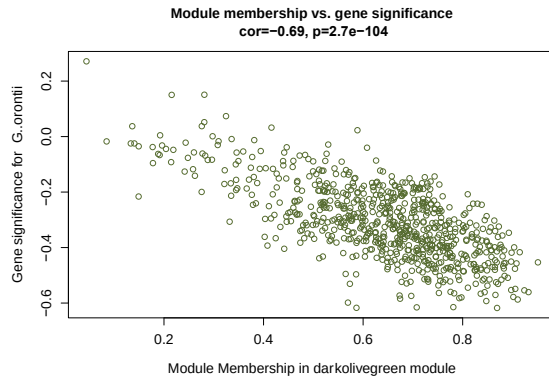
enriched in flavonoid biosynthesis pathway, starch and sucrose metabolism, nicotinate and nicotinamide metabolism

→ Flavonoids as defense mechanism against biotrophic pathogens

Golovinomyces orontii

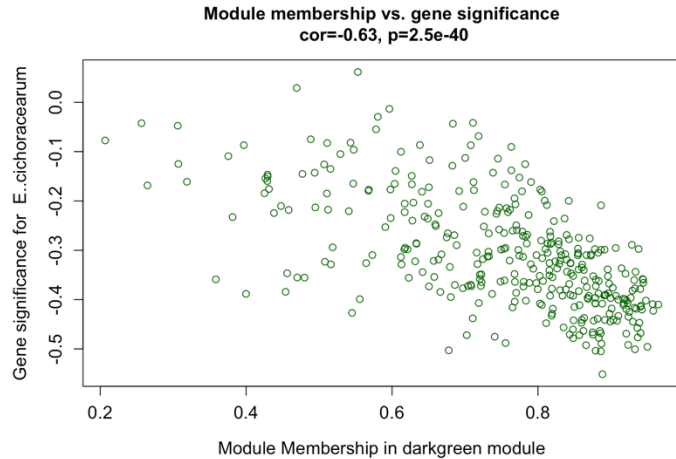


- Downregulation of protein export, protein localization to ER



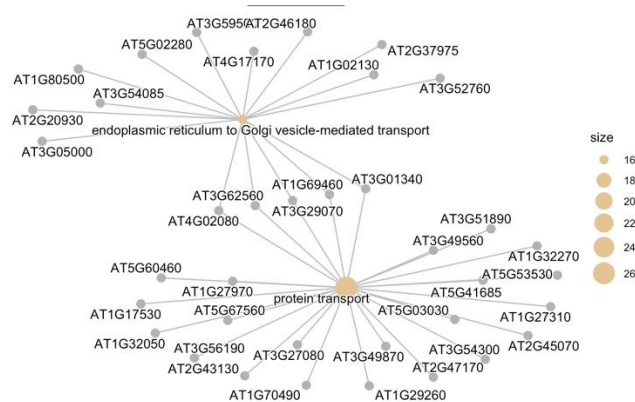
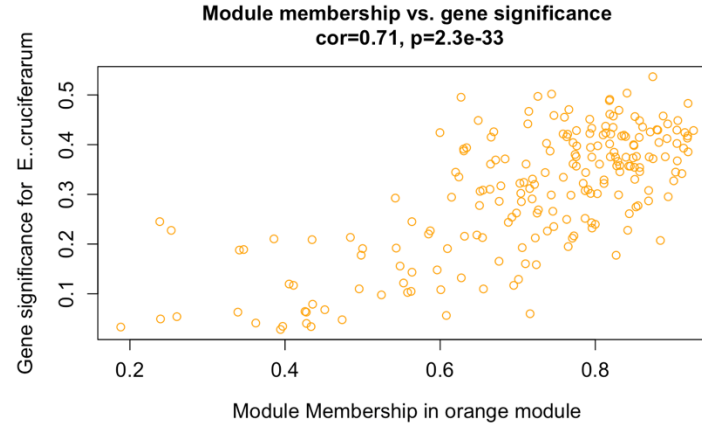
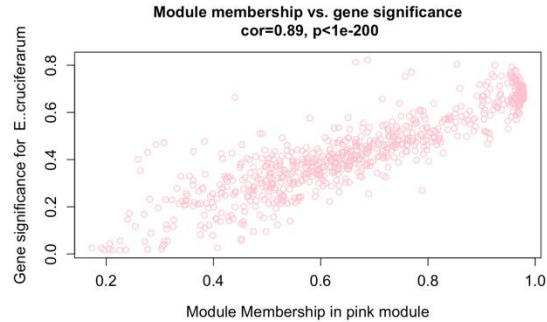
- Response to hypoxia and to carbohydrate

Erysiphe cichoracearum



- Darkgreen module: enriched in Translation (Ribosome)
→ downregulated during infection

Erysiphe cruciferarum



Description <chr>

GeneRatio <chr>

Ribosome – Arabidopsis thaliana (thale cress)

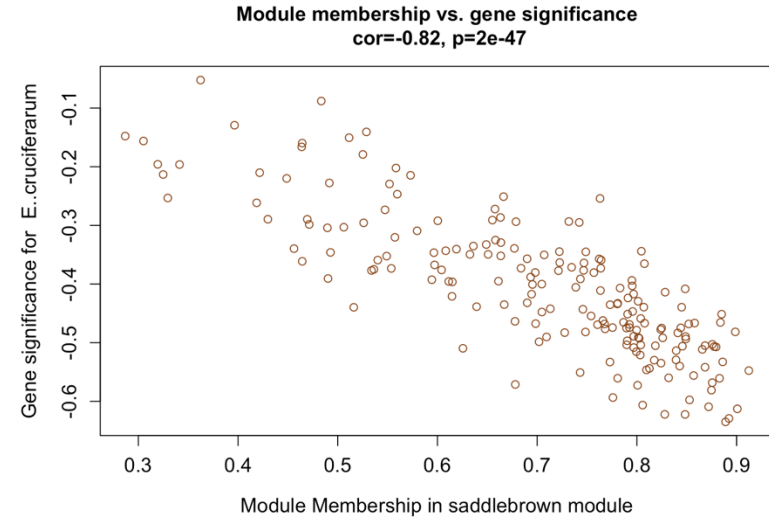
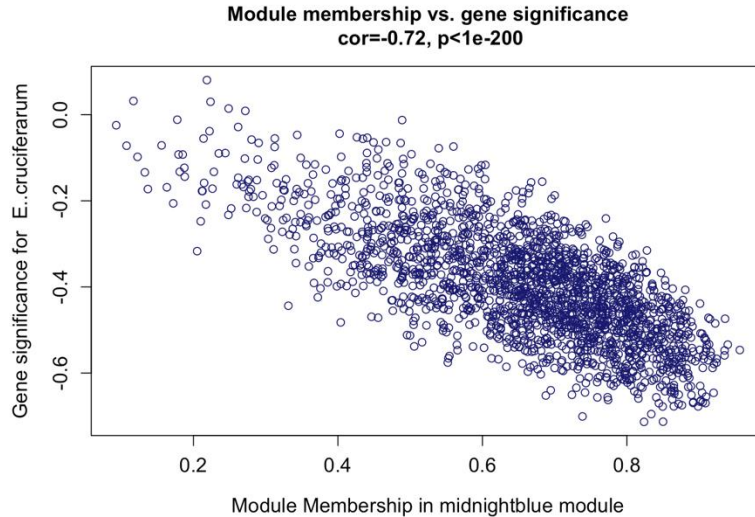
37/57

Proteasome – Arabidopsis thaliana (thale cress)

5/57

- unpublished dataset: *E. cruciferarum*
- Strongly upregulated in pink and orange module

Erysiphe cruciferarum



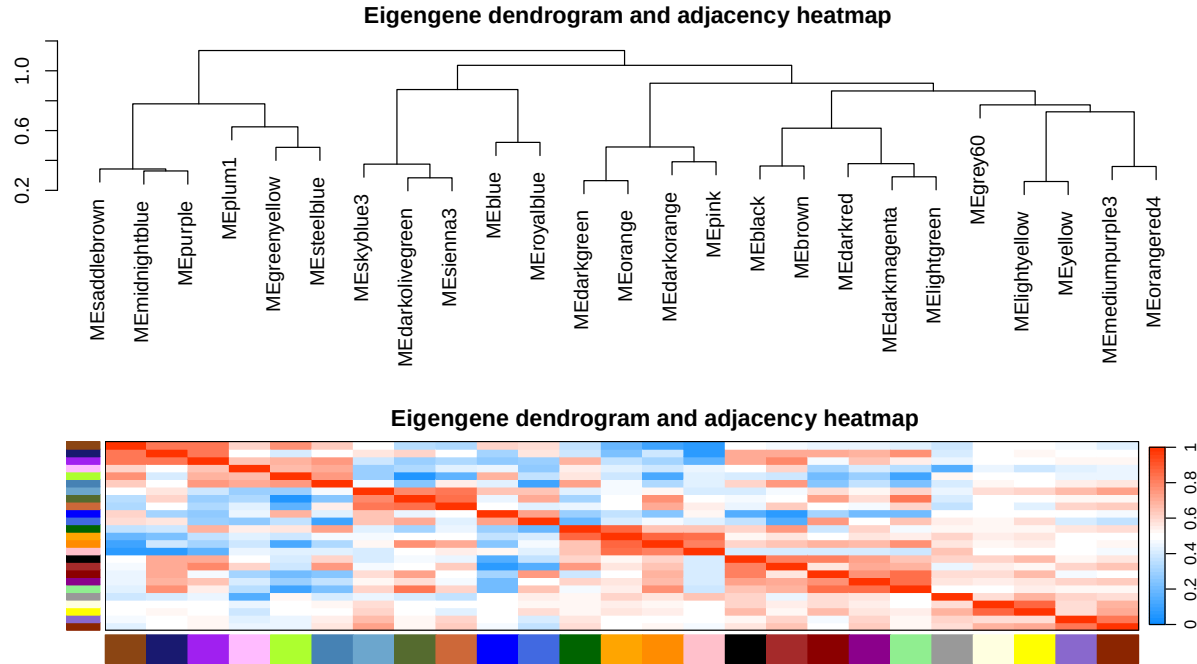
Protein processing in endoplasmic reticulum – *Arabidopsis thaliana* (thale cress)

Nucleocytoplasmic transport – *Arabidopsis thaliana* (thale cress)

Viral life cycle – HIV-1 – *Arabidopsis thaliana* (thale cress)

Ubiquitin mediated proteolysis – *Arabidopsis thaliana* (thale cress)

RNA degradation – *Arabidopsis thaliana* (thale cress)



Commonalities and Divergences in Biotrophic Plant-Microbe Interactions

- Each biotroph seems to have its own strategy
- Manipulation of plant's metabolisms for nutrient uptake
- Some biotrophs suppress plant defense
- Some biotrophs actively use plant defense mechanisms to their advantage
- A lot of different modules could also result from the way the dataset is build